Chunghwan Lee

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Research Interests

- Reconstructing 3D scenes
- Privacy-preserving visual localization
- Neural fields in 3D visual computing

Education

Hanyang University (HYU) Seoul, Korea
M.S., Dept. of Electronic Engineering (*GPA: 4.33/4.5 (until 2ndsem.)*) *Mar. 2022 – Present*
Advisor: Prof. Je Hyeong Hong

Hanyang University (HYU) Seoul, Korea
B.S., Dept. of Electronic Engineering (*GPA: 3.82/4.5*) *Mar. 2015 – Feb. 2022*
Thesis: *Deepfake Detection Using Supervised Temporal Feature Extraction Model and LSTM*

Publications

1. "**Paired-Point Lifting for Enhanced Privacy-Preserving Visual Localization**", **Chunghwan Lee**, Jaihoon Kim, Chanhyuk Yun, Je Hyeong Hong, *The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023*
2. **"Deepfake Detection Using Supervised Temporal Feature Extraction Model and LSTM"**, **Chunghwan Lee(*)**, Jaihoon Kim(*), Kijung Yoon (* *equal contribution), The Korean Institute of Broadcast and Media Engineers Fall Conference, 2022*
3. **"Weather Classification and Image Restoration Algorithm Attentive to Weather Conditions in Autonomous Vehicles"**, Jaihoon Kim(*), **Chunghwan Lee(*)**, Sangmin Kim, Jechang Jeong (* *equal contribution), The Korean Institute of Broadcast and Media Engineers Fall Conference, 2021*

Patents

(Korea) Localization method and apparatus using line cloud map data, line cloud cloud map data generation method, **Chunghwan Lee**, Je Hyeong Hong, Chanhyuk Yun, *Registration No. 10-2023-0008666 (applicant: HYU)* *Jan. 2023*

Research Projects

HY-Vision Lab, Hanyang University (HYU) Seoul, Korea
Enhancing privacy-preserving visual localization *Feb. 2022 – Present*

- Proposed a 3D line cloud generation method for privacy concealment, which is a simple lightweight algorithm, named paired-point lifting (PPL).
- Conducted extensive experiments to confirm that PPL conceals scene details without impairing the accuracy of pose estimation during visual localization.

Deep learning-based Montage Face Recognition *Oct. 2021 – Feb.2022*

- A project to identify a criminal from facial composite (sketched montage); KIST (Seoul, Korea) contributed real-life sketch data.
- Developed human face recognition model by fine-tuning the ARKFACE and conducting an augmentation on montage sketch data with CycleGAN.

NIAI Lab, Hanyang University (HYU) Seoul, Korea
Deepfake Video Detection *Feb. 2021 – Sep. 2021*

- Developed the classification model discerning deepfake videos by training the model to extract the abrupt features.
- Utilized singular vector decomposition (SVD) to gather orthogonal vectors from the feature domain.

ICSP Lab, Hanyang University (HYU) Seoul, Korea

2D image-based weather classification and restoration *Jun. 2020 – Oct. 2020*

- Designed a one-for-all model that processes from classification to reconstruction from bad weather artifacts; Derain, Desnow, Dehaze.
- Trained a classifier and reconstruction models at once; mobilenet-based classifier discerning sunny, rainy, snowy, and hazy weather was trained with the MWI dataset, and each Unet-based reconstruction model was trained to remove one type of weather artifact.
- Increased reconstruction performance by automatically choosing the best model that can remove weather artifacts.

Skills

Programming proficient in: Python (including libraries such as Numpy, Pandas, OpenCV, Open3D), PyTorch, Keras and Tensorflow (including visualization and debugging tools such as Tensorboard and Weights & biases); familiar with: C/C++, Java, MATLAB

Honors

Third Place, IC-PBL+ Graduate Championship 2022-2, IC-PBL Teaching and Learning Center, HYU, Seoul, Korea *2023*

Excellence Award, Undergraduate Hanyang Academic, HYU, Seoul, Korea *2022*

Excellence Award, The Korean Institute of Broadcast and Media Engineers Fall Conference 2021, The KIBME, Seoul, Korea *2021*

Scholarships

HY-in Scholarships, Graduate Hanyang University, Seoul, Korea *2022 – present*
$ 12,200

Dept. E.E. Scholarship, Undergraduate Hanyang University, Seoul, Korea *2015 – 2016*
$ 14,500

Additional

Paper reviewer - ICIP 2022-2023, WACV 2023-2024, ICCV 2023